HARD · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Mixed

02

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 157.8

Accept also: 789/5

The correct answer is 157.8. It's given that an area of 46.00 square nautical miles is equivalent to k square kilometers. Since 1 nautical mile is equal to 1.852 kilometers, it follows that an area of 1 square nautical mile is equivalent to $(1.852)^2$, or 3.429904, square kilometers. Multiplying 46.00 by 3.429904 yields 157.775584 square kilometers. Rounding this value to the nearest tenth yields 157.8. Therefore, the value of k is 157.8.

Q2 D

D. Sample A had a smaller sample size.

Choice D is correct. Sample size is an appropriate reason for the margin of error to change. In general, a smaller sample size increases the margin of error because the sample may be less representative of the whole population.

Choice A is incorrect. The margin of error will depend on the size of the sample of recorded votes, not the number of votes that could not be recorded. In any case, the smaller number of votes that could not be recorded for sample A would tend to decrease, not increase, the comparative size of the margin of error. Choice B is incorrect. Since the percent in favor for sample A is the same distance from 50% as the percent in favor for sample B, the percent of favorable responses doesn't affect the comparative size of the margin of error for the two samples. Choice C is incorrect. If sample A had a larger margin of error than sample B, then sample A would tend to be less representative of the population. Therefore, sample A is not likely to have a larger sample size.

Q3 10

The correct answer is 10. Solving by substitution, the given system of equations, where p is a constant, can be written so that the left-hand side of each equation is equal to $7r$. Subtracting 6 from each side of the first equation in the given system, $6 + 7r = pw$, yields $7r = pw - 6$. Adding $5w$ to each side of the second equation in the given system, $7r - 5w = 5w + 11$, yields $7r = 10w + 11$. Since the left-hand side of each equation is equal to $7r$, setting the the right-hand side of the equations equal to each other yields $pw - 6 = 10w + 11$. A linear equation in one variable, w, has no solution if and only if the equation is false; that is, when there's no value of w that produces a true statement. For the equation $pw - 6 = 10w + 11$, there's no value of w that produces a true statement when $pw = 10w$. Therefore, for the equation $pw - 6 = 10w + 11$, there's no value of w that produces a true statement when the value of p is 10. It follows that in the given system of equations, the system has no solution when the value of p is 10.

Q4 20

The correct answer is 20. For the given circuit, the resistance R is 500 ohms, and the total potential difference V generated by n batteries is $6n$ volts. It's also given that the circuit is to have a current of no more than 0.25 ampere, which can be expressed as $I < 0.25$. Since Ohm's law says that $I = \frac{V}{R}$, the given values for V and R can be substituted for I in this inequality, which yields $\frac{6n}{500} < 0.25$. Multiplying both sides of this inequality by 500 yields $6n < 125$, and dividing both sides of this inequality by 6 yields $n < 20.833$. Since the number of batteries must be a whole number less than 20.833, the greatest number of batteries that can be used in this circuit is 20.

Q5 A

A. $fx = -d - cx$

Choice A is correct. It’s given that the graph of the linear function $y = fx + 19$ is shown. This means that the graph of $y = fx + 19$ can be translated down 19 units to create the graph of $y = fx$ and the y-coordinate of every point on the graph of $y = fx + 19$ can be decreased by 19 to find the resulting point on the graph of $y = fx$. The y-intercept of the graph of $y = fx + 19$ is $0, 3$. Translating the graph of $y = fx + 19$ down 19 units results in a y-intercept of the graph of $y = fx$ at the point $0, 3 - 19$, or $0, -16$. The graph of $y = fx + 19$ slants down from left to right, so the slope of the graph is negative. The translation of a linear graph changes its position, but does not change its slope. It follows that the slope of the graph of $y = fx$ is also negative. The equation of a linear function f can be written in the form $fx = b + mx$, where b is the y-coordinate of the y-intercept and m is the slope of the graph of $y = fx$. It's given that c and d are positive constants. Since the y-coordinate of the y-intercept and the slope of the graph of $y = fx$ are both negative, it follows that $fx = -d - cx$ could define f.

Choice B is incorrect. This could define a linear function where its graph has a positive, not negative, y-intercept.

Choice C is incorrect. This could define a linear function where its graph has a positive, not negative, slope.

Choice D is incorrect. This could define a linear function where its graph has a positive, not negative, y-intercept and a positive, not negative, slope.

Q6

5

The correct answer is 5. The standard form of an equation of a circle in the xy-plane is $x - h^2 + y - k^2 = r^2$, where h, k, and r are constants, the coordinates of the center of the circle are h, k, and the length of the radius of the circle is r. It's given that an equation of the circle is $x - 2^2 + y - 9^2 = r^2$. Therefore, the center of this circle is $2, 9$. It's given that the endpoints of a diameter of the circle are $2, 4$ and $2, 14$. The length of the radius is the distance from the center of the circle to an endpoint of a diameter of the circle, which can be found using the distance formula, $\sqrt{x_1 - x_2{}^2 + y_1 - y_2{}^2}$. Substituting the center of the circle $2, 9$ and one endpoint of the diameter $2, 4$ in this formula gives a distance of $\sqrt{2 - 2^2 + 9 - 4^2}$, or $\sqrt{0^2 + 5^2}$, which is equivalent to 5. Since the distance from the center of the circle to an endpoint of a diameter is 5, the value of r is 5.

Q7

C

C. $x - 11^2 + y + 3^2 = 1$

Choice C is correct. The equation of a circle in the xy-plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r units. It's given that circle A has the equation $(x - 7)^2 + (y + 3)^2 = 1$, which can be written as $(x - 7)^2 + (y - (-3))^2 = 1^2$. It follows that $h = 7$, $k = -3$, and $r = 1$. Therefore, the center of circle A is $(7, -3)$ and its radius is 1 unit. If circle A is translated 4 units to the right, the x-coordinate of the center will increase by 4, while the y-coordinate and the radius of the circle will remain unchanged. Translating the center of circle A to the right 4 units yields $(7 + 4, -3)$, or $(11, -3)$. Therefore, the center of circle B is $(11, -3)$. Substituting 11 for h, -3 for k, and 1 for r into the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 11)^2 + (y - (-3))^2 = 1$, or $(x - 11)^2 + (y + 3)^2 = 1$. Therefore, the equation $(x - 11)^2 + (y + 3)^2 = 1$ represents circle B.

Choice A is incorrect. This equation represents a circle obtained by shifting circle A down, rather than right, 4 units.

Choice B is incorrect. This equation represents a circle obtained by shifting circle A left, rather than right, 4 units.

Choice D is incorrect. This equation represents a circle obtained by shifting circle A up, rather than right, 4 units.

Q8

120

The correct answer is 120. The solutions to a quadratic equation of the form $ax^2 + bx + c = 0$ can be calculated using the quadratic formula and are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 2$, $b = -8$, and $c = -7$. It follows that the solutions to the given equation are $x = \frac{8 \pm \sqrt{-8^2 - 42 - 7}}{22}$, which is equivalent to $x = \frac{8 \pm \sqrt{64 + 56}}{4}$, or $x = \frac{8 \pm \sqrt{120}}{4}$. It's given that one solution to the equation $2x^2 - 8x - 7 = 0$ can be written as $\frac{8 - \sqrt{k}}{4}$. The solution $\frac{8 - \sqrt{120}}{4}$ is in this form. Therefore, the value of k is 120.

Q9

D

D. -25

Choice D is correct. Substituting b for x in the given equation yields $g\left(b + 28\right) = \frac{1}{10}\left(b - a\right)^2\left(b - b\right)^2\left(b - c\right)$, which is equivalent to $g\left(b + 28\right) = \frac{1}{10}\left(b - a\right)^2\left(0\right)^2\left(b - c\right)$, or $g\left(b + 28\right) = 0$. It follows that when $x = b + 28$, $g\left(x\right) = 0$. There are three points where the graph of $y = g\left(x\right)$ intersects the x-axis: $\left(-2, 0\right)$, $\left(1, 0\right)$, and $\left(3, 0\right)$. Therefore, $g\left(-2\right) = 0$, $g\left(1\right) = 0$, and $g\left(3\right) = 0$. It follows that either $b + 28 = -2$, $b + 28 = 1$, or $b + 28 = 3$. Subtracting 28 from each side of the equation $b + 28 = -2$ yields $b = -30$. Subtracting 28 from each side of the equation $b + 28 = 1$ yields $b = -27$. Subtracting 28 from each side of the equation $b + 28 = 3$ yields $b = -25$. Thus, the value of b must be either -30, -27, or -25. Of these values, only -25 is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This would be a possible value of b if the graph shown was the graph of $y = g\left(x + 28\right)$, not $y = g\left(x\right)$.

Choice C is incorrect. This would be the value of c, not the value of b, if the graph shown was the graph of $y = g\left(x + 28\right)$, not $y = g\left(x\right)$.

Q10 18

The correct answer is 18. Subtracting 45 from each side of the given equation yields $x - 9 = 18$. By the definition of absolute value, if $x - 9 = 18$, then $x - 9 = 18$ or $x - 9 = -18$. Adding 9 to each side of the equation $x - 9 = 18$ yields $x = 27$. Adding 9 to each side of the equation $x - 9 = -18$ yields $x = -9$. Therefore, the solutions to the given equation are 27 and -9, and it follows that the sum of the solutions to the given equation is $27 + -9$, or 18.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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